Viral Diagnosis: How PCR Finds a Virus

From a swab to a result — copying tiny amounts of genetic material until we can detect them

1. Why do we need tests to find viruses?

A doctor cannot tell from symptoms alone which virus has made a person ill — many viruses cause fever and cough. To be sure, a clinical sample (such as a throat swab or blood) is tested in a laboratory for the virus itself, for viral parts (antigens) or for the antibodies the patient's body made against it.

Modern tests have changed a great deal. Very sensitive antibody tests and nucleic-acid tests have mostly replaced the old method of growing the virus in cells. A good diagnostic test should be fast, simple, sensitive (does not miss real cases), specific (does not give false alarms) and affordable.

Test	What it looks for	Speed	Sensitivity	Notes
Virus culture	Live virus grown in cells	Days to weeks	Moderate	Slow, needs special labs; rarely used for routine diagnosis
Antigen test (e.g. rapid test strip)	Viral proteins	Minutes	Lower	Cheap and simple; best when the person has a lot of virus
Antibody test (EIA/ELISA)	The body's antibodies	Hours	High	Shows past or recent infection, not always current
PCR / RT-PCR	Viral genetic material (DNA or RNA)	Hours	Very high	The "gold standard" for many viruses; needs equipment

2. Viruses in brief

Viruses are tiny infectious particles that can only multiply inside the living cells of another organism. They can infect every kind of life — plants, animals, bacteria (bacteriophages) and archaea. A virus carries its genetic material, either DNA or RNA, inside a protein coat. A virus causes disease when it enters the cells of a host and takes over their machinery to make more viruses.

3. The polymerase chain reaction (PCR)

PCR is a laboratory technique for making millions or billions of copies of one chosen region of DNA. The region can be anything the scientist is interested in: a gene to be studied, a genetic marker used in forensics, or a short stretch of a virus's genome. It was conceived by Kary Mullis in 1983 (Nobel Prize in Chemistry, 1993) and has transformed medicine and biology because it lets us detect even a handful of DNA molecules.

PCR copies DNA the same way a cell does (see Day 1). It needs a DNA polymerase that builds new strands using an existing strand as a template. The enzyme used is Taq polymerase, from the heat-loving bacterium Thermus aquaticus, which lives in hot springs. Taq works best at about 72 °C and survives the repeated heating to 95 °C that would destroy a human or E. coli polymerase. This heat-stability is what makes automatic PCR possible.

The ingredients

Ingredient	Role
Template DNA	The sample containing the region to be copied
Two primers	Short single-stranded pieces of DNA (about 18–25 bases). They are designed to match the two edges of the target region, one on each strand. They give the polymerase a starting point.
Taq polymerase	Builds the new DNA strands
Nucleotides (dNTPs)	The A, T, G, C building blocks
Buffer and magnesium	The right chemical conditions for the enzyme

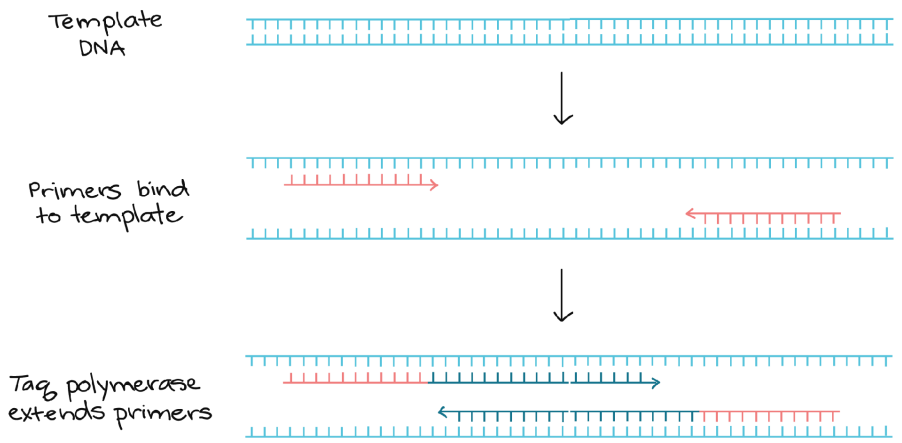


Figure 1. Top to bottom: template DNA; primers bind to the template (red arrows); Taq polymerase extends them to copy the region between. *Source: Khan Academy*

Primers are what give PCR its specificity: they only bind where their sequence matches, so only the target is copied. The primers bind the template by complementary base pairing, exactly as you learned for DNA.

The three steps of one cycle

Step	Temperature	What happens
1. Denaturation	about 95 °C	Heat separates the two strands of the DNA. This gives single-stranded templates.
2. Annealing	55–65 °C	The mixture is cooled so the primers can bind to their matching sequences.

Step	Temperature	What happens
3. Extension	72 °C	Taq polymerase extends the primers and builds new strands.

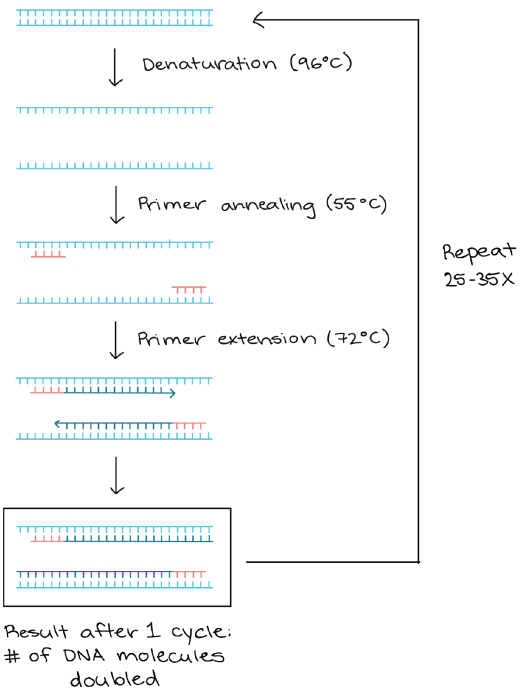


Figure 2. One PCR cycle drawn as a flowchart: denaturation at 96 °C, primer annealing, then primer extension. The cycle is repeated 25–35 times. After one cycle the number of DNA molecules has doubled. *Source: Khan Academy*

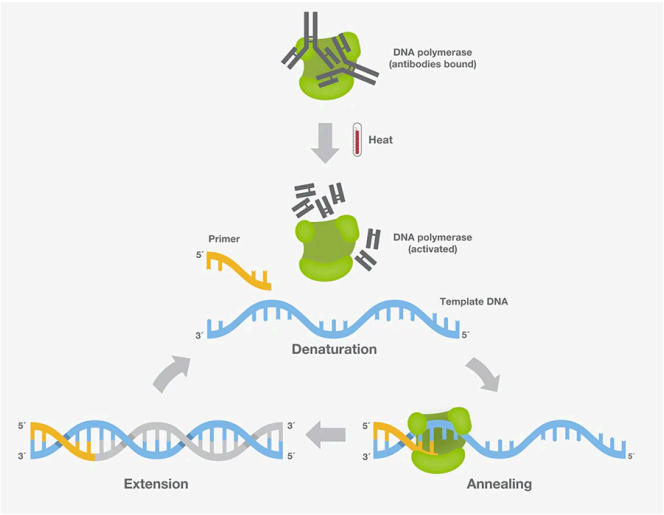


Figure 3. The same three steps drawn around the enzyme. (In this version antibodies hold the polymerase inactive until the first heating — a "hot-start" PCR that improves accuracy.) *Source: Class material*

Why does it make billions of copies?

A typical PCR repeats the cycle 25–35 times, which takes roughly 1–3 hours depending on the length of the region. The new DNA made in one cycle becomes a template in the next one, and plenty of primers and enzyme are present, so the number of DNA molecules doubles with every cycle.

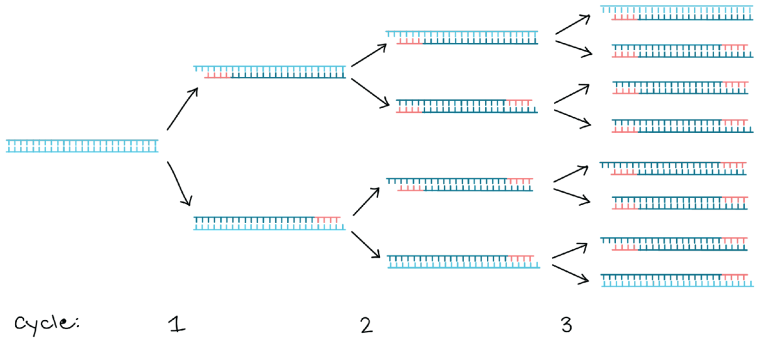


Figure 4. Each cycle doubles the number of copies of the target region. *Source: Khan Academy*

Cycles	Copies (from 1 starting molecule)
0	1
1	2
2	4
3	8
10	1,024
20	about 1 million (1,048,576)
30	about 1 billion (1,073,741,824)

IN REAL LIFE
Doubling needs a perfect reaction. In practice each cycle is slightly less than 100 % efficient, and the reaction plateaus when ingredients run out.

ACTIVITY: THE DOUBLING GAME (5 MINUTES)
Each student starts with one bead. At each "cycle" every student doubles their beads (take from a central pot). How many cycles until the pot — and the room — is empty? Then ask: how many cycles would be needed to reach a billion?

4. Detecting RNA viruses: RT-PCR and real-time PCR

Many dangerous viruses, such as SARS-CoV-2, the flu virus and HIV, have RNA genomes, but Taq polymerase copies DNA only. So the first step is to convert the RNA to DNA using the enzyme reverse transcriptase, which makes a cDNA copy. PCR on that cDNA is called RT-PCR.

Real-time (quantitative) PCR adds a fluorescent dye or probe that glows more brightly as more DNA is made. A machine measures the light after every cycle. The cycle at which the glow rises above a threshold is the Ct value (cycle threshold).

HOW TO READ A CT VALUE

- **Low Ct** = the target appeared early = there was **a lot** of virus in the sample (high viral load).
- **High Ct** = the target appeared late = there was **little** virus.
- No signal by about cycle 35–40 = **negative** or non-diagnostic. Cut-off values differ between tests and laboratories.

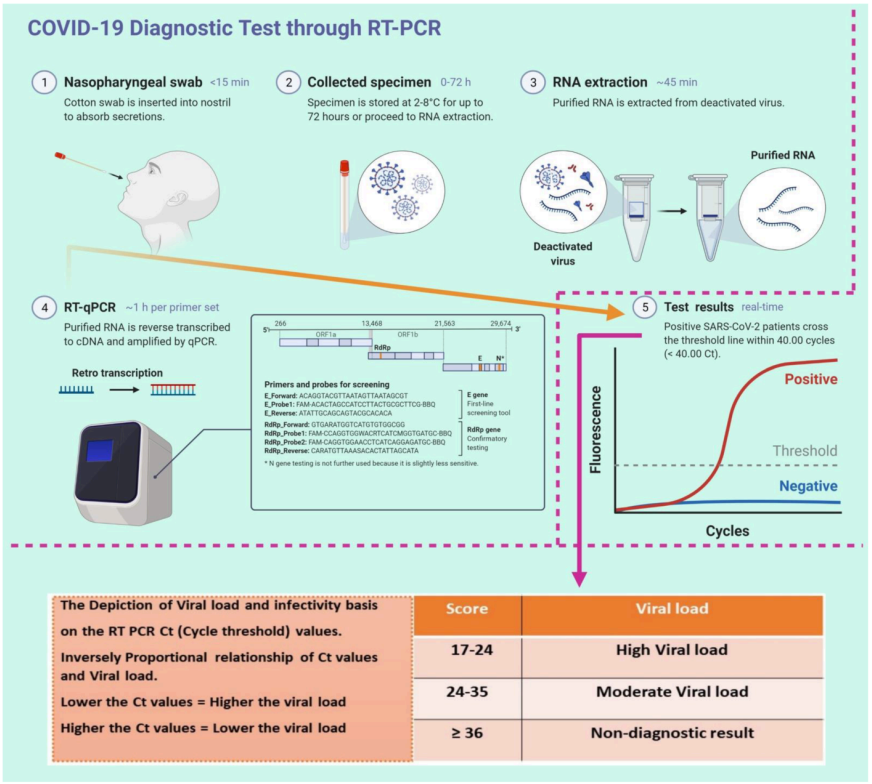


Figure 5. The RT-PCR workflow for COVID-19: swab, specimen, RNA extraction, RT-qPCR and results. The table at the bottom shows how Ct values relate to viral load in this example. *Source: Diagnostics (MDPI)*

Test your understanding

These are the Ct values from RT-PCR tests of five patients suspected of having SARS-CoV-2:

Patient	a	b	c	d	e
Ct value	23	30	18	20	37

1. Rank the patients from **highest to lowest viral load**.
2. Who do you think is **most severely affected**? What other information would you need to be sure?

The answers are in the Quizzes and Answer Keys document.

5. Gel electrophoresis: seeing the DNA

After PCR we often need to check that DNA of the right size was made. Gel electrophoresis does this. DNA is a negatively charged molecule (its phosphate backbone carries a negative charge), so in an electric field it travels toward the positive electrode.

1. A jelly slab of **agarose** is made with small wells at one end.
2. DNA samples, plus a **DNA ladder** (fragments of known size), are loaded into the wells.
3. A current is switched on. DNA moves through the gel's tiny pores toward the positive end. **Small fragments move faster and farther; large fragments move slower.**
4. A stain that binds DNA (and glows under special light) shows the DNA as **bands**. Compare each band with the ladder to estimate its size.

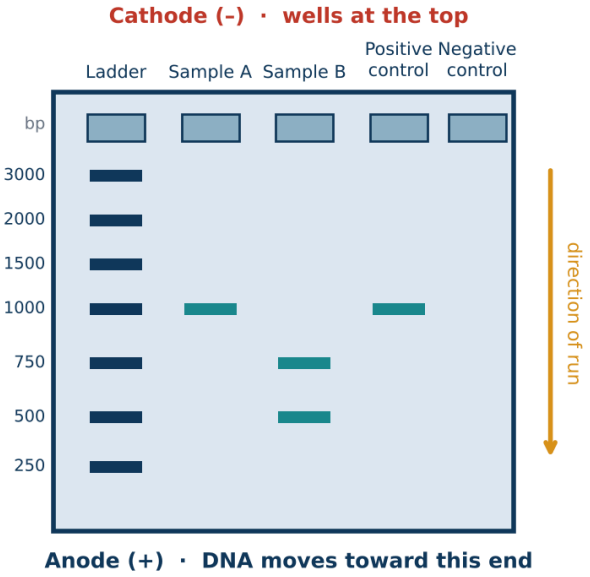


Figure 6. A schematic gel. Sample A makes a single band of about 1,000 bp, like the positive control; sample B shows two smaller fragments; the negative control has no band. *Source: Own illustration*

READING THE GEL

The positive control shows the test works; the negative control shows no contamination. Without both, the result cannot be trusted.

Check your understanding

1. Why does PCR use a heat-stable polymerase?
2. What is the job of the primers?
3. After 25 perfect cycles, how many copies would come from one molecule?
4. Why is RT-PCR needed to detect SARS-CoV-2?
5. In which direction does DNA move in a gel, and why?